

## Text Files: Opening

You use **fopen** to open a file. It opens a file for a specified mode (the three most common are r, w, and a, for read, write, and append). It then returns a file pointer that you use to access the file. For example, suppose you want to open a file and write the numbers 1 to 10 in it. You could use the following code:

```
#include <stdio.h>
#define MAX 10

int main()
{
    FILE *f;
    int x;
    f=fopen("out", "w");
    if (!f)
        return 1;
    for(x=1; x<=MAX; x++)
        fprintf(f, "%d\n", x);
    fclose(f);
    return 0;
}
```

The **fopen** statement here opens a file named **out** with the w mode. This is a destructive write mode, which means that if **out** does not exist it is created, but if it does exist it is destroyed and a new file is created in its place. The **fopen** command returns a pointer to the file, which is stored in the variable **f**. This variable is used to refer to the file. If the file cannot be opened for some reason, **f** will contain **NULL**.

### Main Function Return Values

This program is the first program in this series that returns an error value from the main program. If the **fopen** command fails, **f** will contain a **NULL** value (a zero). We test for that error with the **if** statement. The **if** statement looks at the True/False value of the variable **f**. Remember that in C, 0 is False and anything else is true. So if there were an error opening the file, **f** would contain zero, which is False. The **!** is the NOT operator. It inverts a Boolean value. So the **if** statement could have been written like this:

```
if (f == 0)
```

That is equivalent. However, **if (!f)** is more common.

If there is a file error, we return a 1 from the main function. In UNIX, you can actually test for this value on the command line. See the shell documentation for details.

The **fprintf** statement should look very familiar: It is just like **printf** but uses the file pointer as its first parameter. The **fclose** statement closes the file when you are done.